EDA Report

Synthetic Food Delivery Transactions

Exploratory Data Analysis | Python | Pandas | Seaborn | Matplotlib

Dataset: 10,000 Orders | 31 Columns | 2 Years | 8 US Cities

Source: Kaggle

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1. Project Overview

This report presents the findings from an Exploratory Data Analysis (EDA) performed on a synthetic food delivery dataset obtained from Kaggle. The dataset contains 10,000 order records spanning 2 years across 8 major US cities. The analysis covers customer behaviour, delivery performance, revenue patterns, restaurant ratings, and the impact of weather on delivery time.

The objective of this EDA is to identify patterns, trends, and business insights that can help a food delivery platform improve its operations, customer experience, and revenue strategy.

2. Dataset Summary

Property	Details
Source	Kaggle
Total Rows	10,000
Total Columns	31
Time Period	2 Years
Cities	New York, Los Angeles, Chicago, Houston, Phoenix, San Antonio, San Diego, Philadelphia
Order Statuses	Completed, Cancelled
Cuisines	American, Chinese, Italian, Indian, Japanese, Mexican, Mediterranean, Fast Food

3. Data Cleaning and Feature Engineering

Data Cleaning Steps

Step	Action Taken
Datetime Conversion	3 columns (customer_signup_date, order_timestamp, delivery_timestamp) converted from text to datetime
ID Column Conversion	4 ID columns converted from int64 to string as they are labels, not quantities
Null Values - Discount	discount_code and discount_type nulls filled with 'None' as null means no discount applied
Null Values - Event Flag	Created has_event binary column (0/1). Null means no event occurred
Cancelled Orders	775 cancelled orders identified via null delivery_timestamp and filtered into df_delivered
Completed Orders	9,225 completed orders retained in df_delivered for delivery analysis

Feature Engineering

New Column	Description
order_hour	Hour extracted from order_timestamp (0-23)
order_month	Month name extracted from order_timestamp
Time_of_Day	Morning (5-12), Afternoon (12-16), Evening (16-21), Night (21-5)
has_discount	Binary flag: 1 if discount was applied, 0 otherwise
has_event	Binary flag: 1 if a special event occurred on that day, 0 otherwise
signup_year	Year extracted from customer_signup_date

4. Key Metrics

The following table summarises the core business metrics across all 10,000 orders.

Metric	Value	Note
Average Order Value	$35.29	Range: $2.48 to $176.50
Average Delivery Time	48.25 mins	For completed orders only
Average Tip	$3.46	Range: $0.00 to $33.22
Average Delivery Fee	$5.02	Consistent across cities
Average Item Price	$20.14	Per item ordered
Completion Rate	92.25%	9,225 of 10,000 orders completed
Cancellation Rate	7.75%	775 orders cancelled

5. Customer Behaviour Analysis

5.1 Order Volume by Time of Day

Orders were categorised into four time slots based on the hour of placement. Night accounts for the highest order volume followed by Morning. Afternoon is the quietest period, likely because customers are at work or busy.

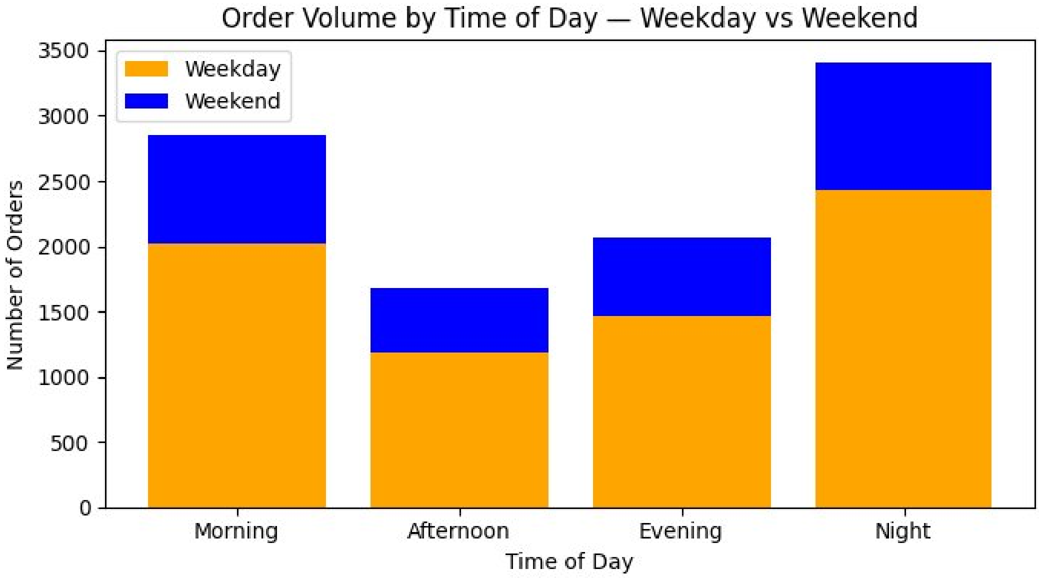

Figure 1: Order Volume by Time of Day — Weekday vs Weekend

> ***Key Finding:*** *Night orders (3,407) are the highest across both weekdays and weekends. The weekday-to-weekend ratio remains consistent across all time slots, indicating no major behavioural shift on weekends.*

5.2 Order Volume by Hour of Day

Order volume is distributed almost uniformly across all 24 hours. No single hour shows a significant spike. This indicates the platform receives consistent demand throughout the day.

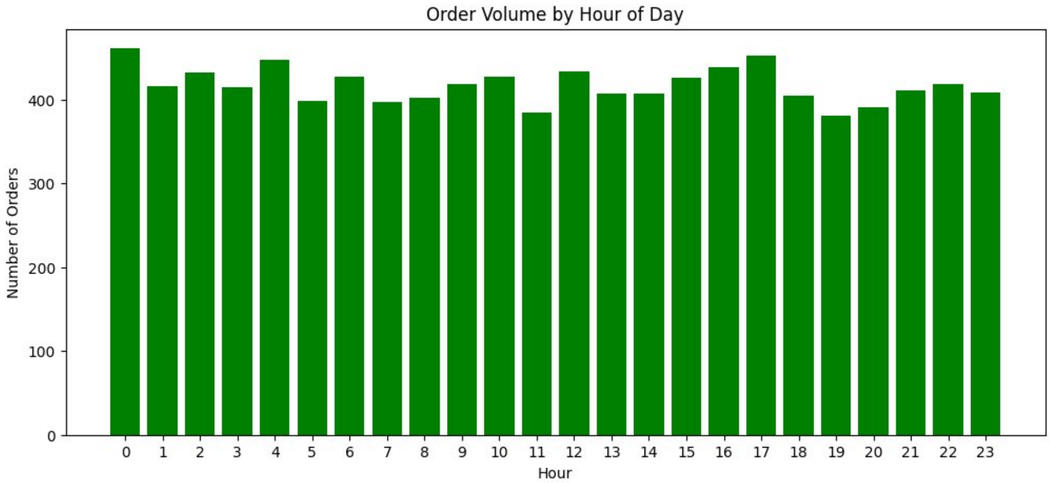

Figure 2: Order Volume by Hour of Day

Key Finding: *Hours 0 and 17 show slightly higher volumes but the overall distribution is flat, confirming that the platform operates with steady demand at all hours.*

5.3 Order Volume by City

New York leads with the highest number of orders (1,705), followed by Chicago, San Diego, and Houston. Phoenix has the lowest order volume (752), which is likely due to population size differences across cities.

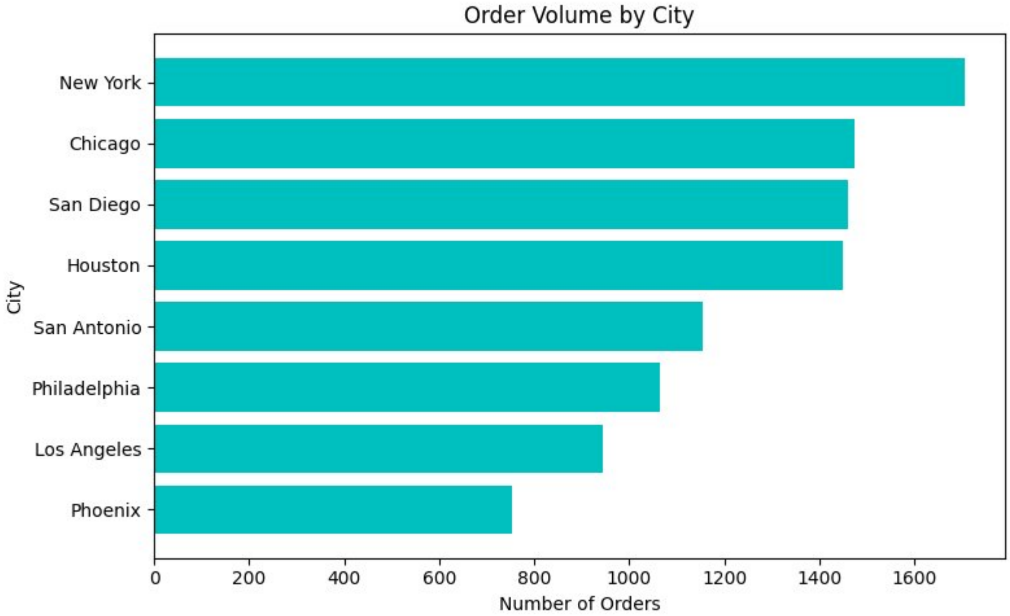

Figure 3: Order Volume by City

> ***Key Finding:*** *New York generates more than double the orders of Phoenix. Cities with lower order volumes like Phoenix also tend to have slower delivery times.*

5.4 Order Value by Loyalty Tier

The violin plot shows the distribution of order value across three loyalty tiers — Gold, Bronze, and Silver. All three tiers have a similar median order value of around $30 and a similar spread, indicating that loyalty tier does not significantly influence spending.

Figure 4: Order Value Distribution by Loyalty Tier

Key Finding: *Gold tier customers do not spend more than Bronze tier customers on average. This is an unexpected finding — loyalty tier is not a reliable indicator of order value.*

6. Delivery Performance Analysis

6.1 Delivery Duration by City

The boxplot shows the spread of actual delivery times across all 8 cities. Los Angeles has the lowest median delivery time, making it the fastest city. Phoenix and Philadelphia show higher medians and wider spreads.

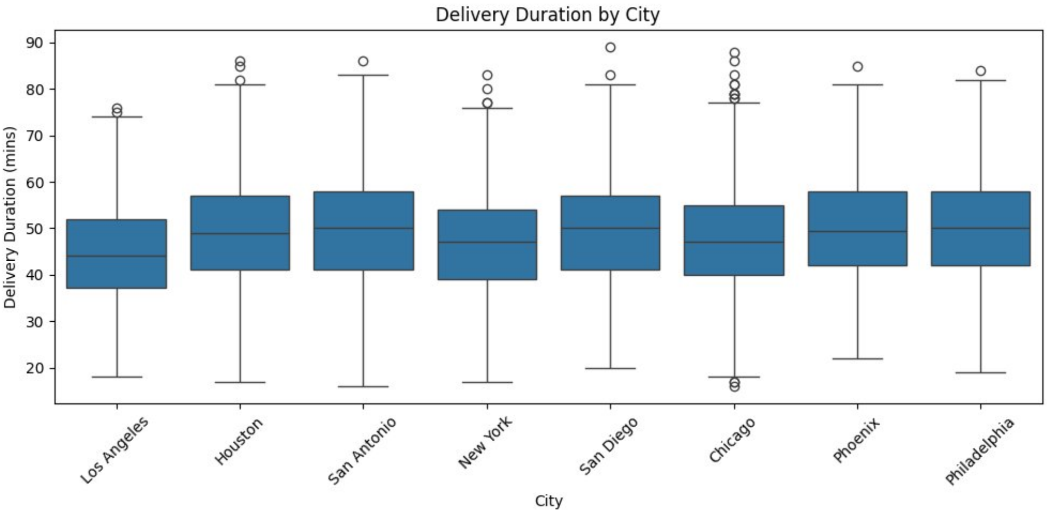

Figure 5: Delivery Duration by City

Key Finding: *Phoenix averages 50.05 minutes for delivery, which is approximately 6 minutes slower than Los Angeles at 44.73 minutes. Cities with fewer orders do not necessarily deliver faster.*

6.2 Impact of Weather on Delivery Time

The scatter plot shows the relationship between weather precipitation and actual delivery duration across all 8 cities. A moderate positive relationship exists (correlation = 0.239), meaning higher rainfall is associated with slightly longer delivery times. However, most orders occur during dry conditions (precipitation near 0).

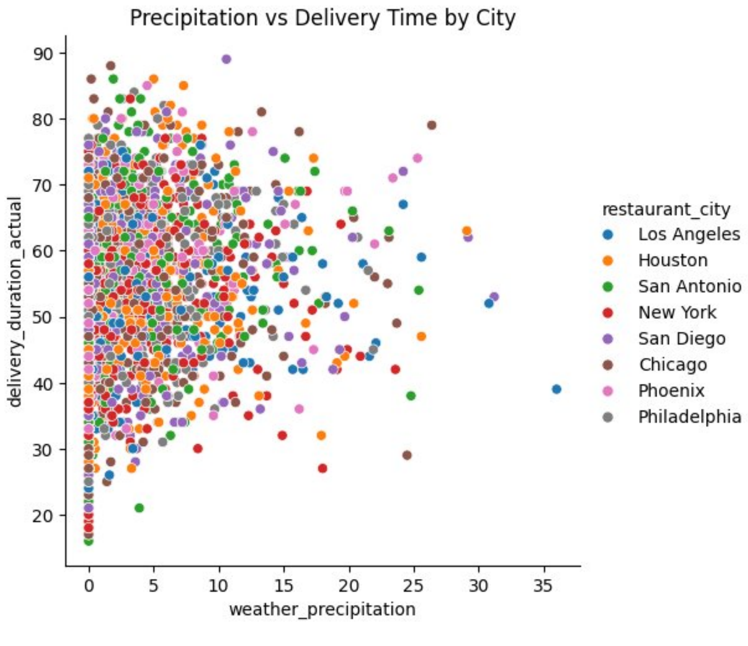

Figure 11: Precipitation vs Delivery Time by City

Key Finding: *Rain moderately increases delivery time but is not the primary driver. The wide spread of delivery times at zero precipitation suggests other factors such as city, prep time, and distance have a greater influence.*

6.3 Distribution of Delivery Duration

The histogram shows that delivery duration follows a roughly bell-shaped distribution centred around 48 minutes. Very few orders fall below 20 minutes or above 75 minutes, indicating consistent and predictable delivery operations.

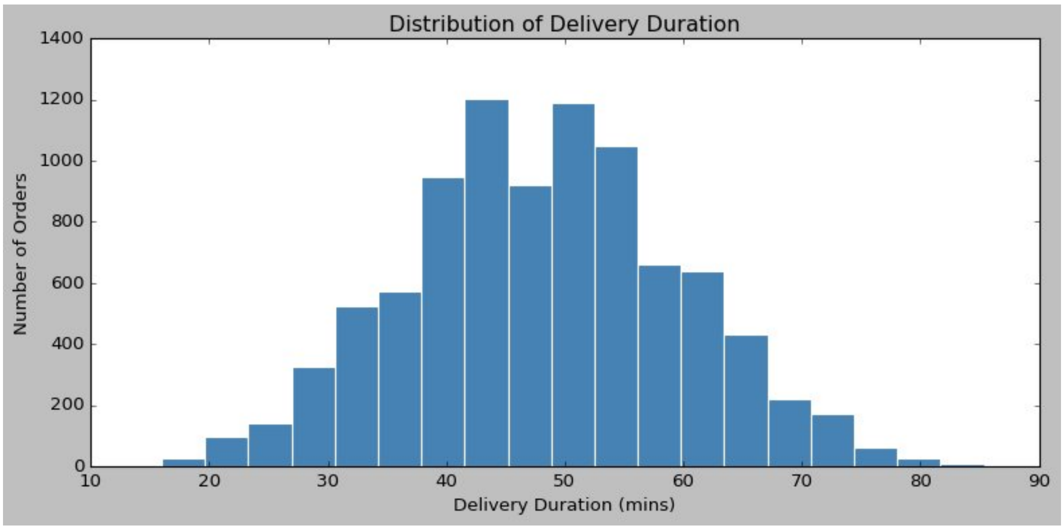

Figure 10: Distribution of Delivery Duration

> ***Key Finding:*** *The delivery duration distribution is consistent and bell-shaped. Only 21 orders exceeded 80 minutes, confirming that extreme delays are rare.*

7. Revenue and Pricing Analysis

7.1 Order Share by Acquisition Channel

All five acquisition channels — paid ads, email, social media, referral, and organic — contribute almost equally to total order volume, each accounting for approximately 20% of orders. No single channel dominates customer acquisition.

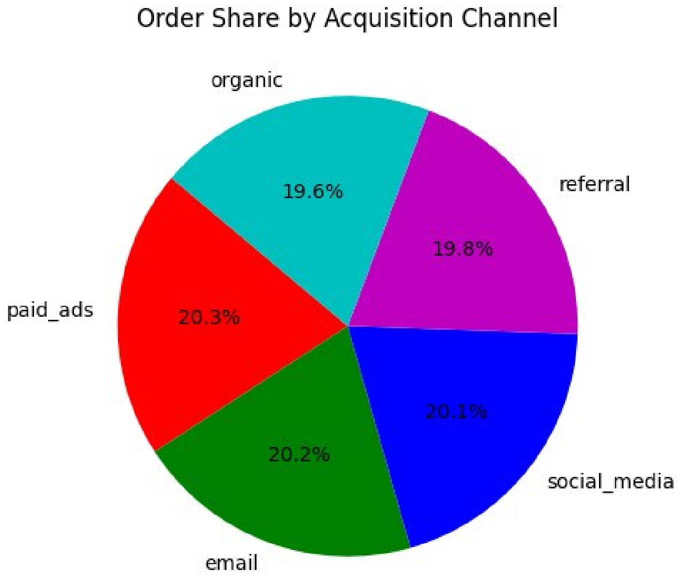

Figure 7: Order Share by Acquisition Channel

> ***Key Finding:*** *The platform has a well-diversified marketing strategy with no over-reliance on any single channel. Email brings in the highest value customers at an average of $35.94 per order.*

7.2 Average Order Value by Month

The line chart shows average order value across all 12 months. The values remain relatively flat throughout the year with no strong seasonal pattern. June is the highest month ($36.41) and February the lowest ($33.90).

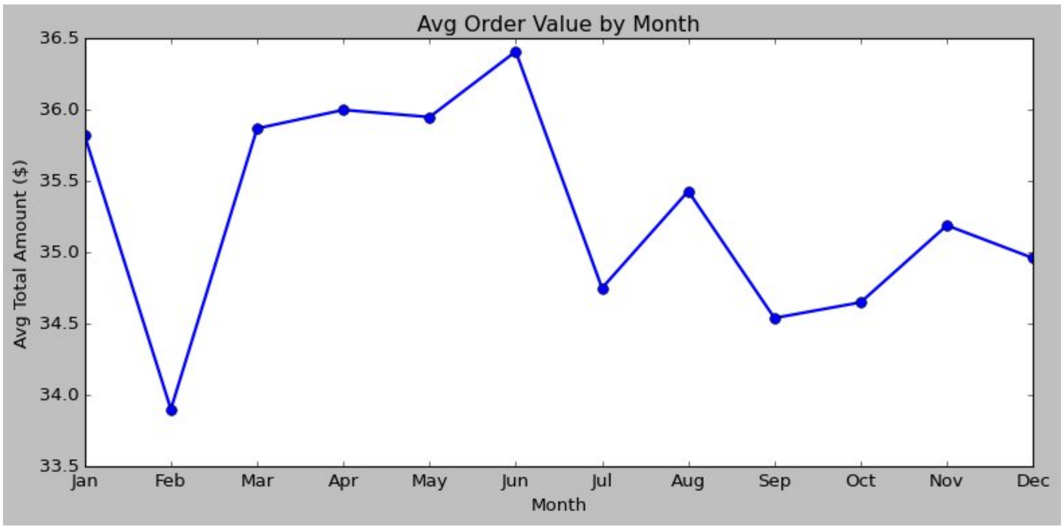

Figure 9: Average Order Value by Month

Key Finding: *Revenue is stable across all months with a difference of only $2.51 between the best and worst performing months. The platform does not show strong seasonal spikes.*

8. Ratings and Satisfaction Analysis

8.1 Average Restaurant Rating by Cuisine

Chinese restaurants have the highest average rating at 4.29, followed by American and Italian. Mexican restaurants are the lowest rated at 3.77 despite having strong order volume. The overall rating range is narrow (3.77 to 4.29), indicating all cuisines are performing reasonably well.

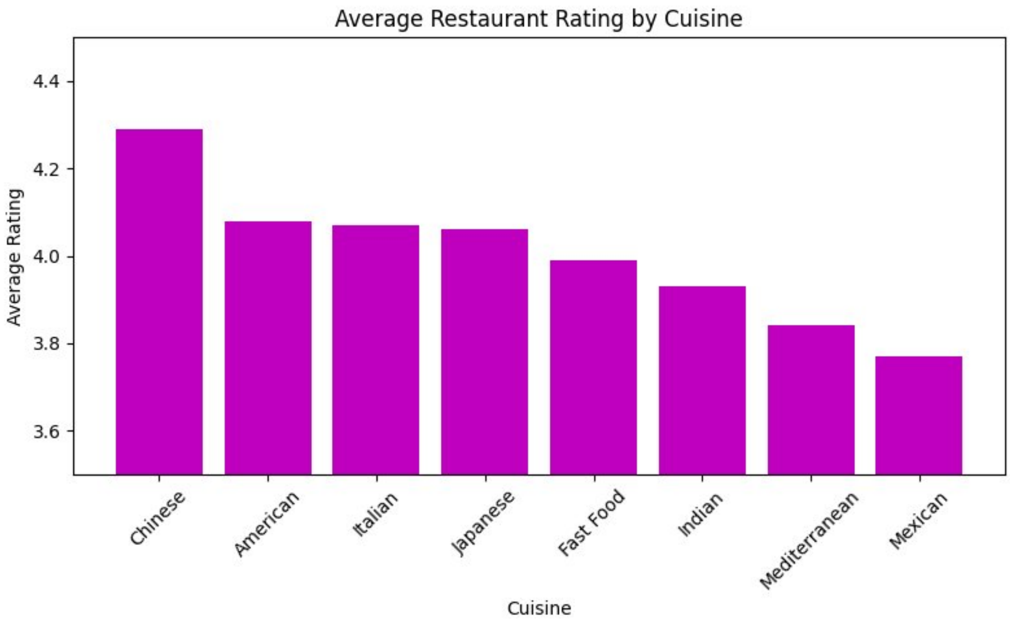

Figure 6: Average Restaurant Rating by Cuisine

> ***Key Finding:*** *Chinese cuisine has the highest rating (4.29) but Indian cuisine generates the highest average order value ($36.94). High spending does not always correlate with high ratings.*

8.2 Order Status Distribution

92.2% of orders were successfully completed and only 7.8% were cancelled. This is a healthy completion rate indicating reliable platform operations.

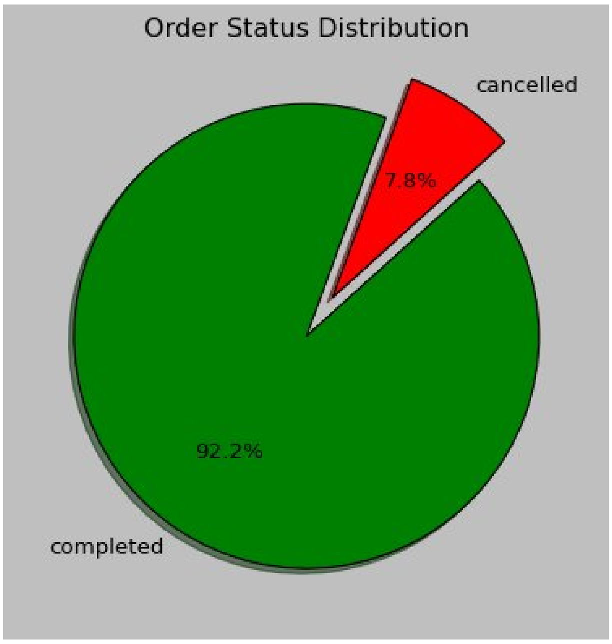

Figure 8: Order Status Distribution

Key Finding: *The 92.2% completion rate reflects stable operations. All 775 cancelled orders had null delivery timestamps, confirming they were never fulfilled.*

9. Visualizations Summary

The following table lists all 11 visualizations created during this EDA, along with the chart type and the key finding from each.

Fig	Title	Chart Type	Key Finding
1	Order Volume by Time of Day	Stacked Bar	Night has the most orders
2	Order Volume by Hour of Day	Bar Chart	Uniform across all 24 hours
3	Order Volume by City	Horizontal Bar	New York leads, Phoenix lowest
4	Order Value by Loyalty Tier	Violin Plot	All tiers spend similarly (~$30 median)
5	Delivery Duration by City	Box Plot	LA fastest, Phoenix slowest
6	Avg Restaurant Rating by Cuisine	Bar Chart	Chinese highest, Mexican lowest
7	Order Share by Acquisition Channel	Pie Chart	All 5 channels equal at ~20%
8	Order Status Distribution	Pie Chart	92.2% completed, 7.8% cancelled
9	Avg Order Value by Month	Line Chart	Flat trend, no seasonal pattern
10	Distribution of Delivery Duration	Histogram	Bell-shaped, centred at 48 mins
11	Precipitation vs Delivery Time	Scatter Plot	Weak positive correlation (0.239)

10. Conclusion and Recommendations

Summary of Findings

Area	Finding
Order Volume	Night is the peak ordering time. Order volume is uniform across all 24 hours.
City Performance	New York has the most orders. Phoenix has the fewest orders and the slowest delivery.
Loyalty Tier	Gold tier customers do not spend more than Bronze tier. Loyalty tier does not drive order value.
Weekend vs Weekday	Weekend orders average $35.77 vs $35.09 on weekdays — a negligible difference.
Delivery Time	Average delivery time is 48.25 minutes. Los Angeles is fastest, Phoenix is slowest.
Weather Impact	Rain slightly increases delivery time (correlation 0.239) but is not the primary cause.
Discounts	Orders with discounts average $31.73 vs $36.15 without — discounts reduce order value.
Ratings	Chinese cuisine is rated highest (4.29). Mexican is lowest (3.77) despite good order volume.
Channels	All 5 acquisition channels contribute equally (~20%). Email brings highest value customers.
Monthly Trend	Order value is flat across all months. No seasonal spikes. June is the best month.
Completion Rate	92.25% of orders are completed. Cancellation rate is 7.75%.

Recommendations

Priority	Area	Recommendation
1	Delivery Speed	Investigate why Phoenix has the slowest delivery despite the lowest order volume. Address operational inefficiencies in that city.
2	Loyalty Program	Review the loyalty program structure. Gold tier customers should be incentivised to spend more than Bronze tier customers.

3	Discount Strategy	Discounts are attracting lower value orders. Review discount targeting to ensure they drive higher spending, not lower.
4	Mexican Cuisine	Mexican restaurants are rated the lowest despite strong order volume. Work with restaurant partners to improve food quality and consistency.

Tools and Libraries Used

Tool / Library	Purpose
Python 3.x	Primary programming language
Pandas	Data loading, cleaning, and analysis
NumPy	Numerical operations
Matplotlib	Bar charts, pie charts, line charts, histograms
Seaborn	Boxplots, violin plots, scatter plots
Jupyter Notebook	Development and documentation environment